

LE1_Review_Questions-Mechanical_Waves

October 8, 2026

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[3]: ## Setup!  
import math
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3. One end of a taut string is shaken at frequency f while the tension is kept constant. The frequency f is then doubled (i.e. $f' = 2f$). What happens to the wave speed v and wavelength λ on the string?

$$v = \lambda 2f \quad (1)$$

$$\lambda = \frac{v}{2f} \quad (2)$$

The wave speed v is directly proportional to the frequency f , so $2f$ results in a doubled wave speed. However, λ is indirectly proportional, so it gets halved.

12. The amplitude A of a wave on a string is doubled while the frequency f , tension F , and the string itself are unchanged. What happens to the average power carried by the wave?

$$P_{av} = \frac{1}{2} \sqrt{\mu F} \times (\omega^2 \times 2A^2) \quad (3)$$

13. Which change would *lower* the pitch (i.e. fundamental frequency) of a vibrating guitar string?

Replacing it with a thicker string of greater mass-per-unit-length μ , but with the same length L and tension F . Recall the formula for *Wave speed on a string*:

$$v = \sqrt{\frac{F}{\mu}} \quad (4)$$

Notice how μ is indirectly related.

14. A wave on a rope has a frequency $f = 5.0$ Hz and a wavelength $\lambda = 0.60$ m. What is the wave speed v ?

$$v = \lambda f \quad (5)$$

$$= 0.60 \text{ m} \times 5.0 \frac{1}{\text{s}} \quad (6)$$

$$= 3.0 \frac{\text{m}}{\text{s}} \quad (7)$$

```
[1]: lamb = 0.60 ## m
    f = 5.0 ## Hz
    v = lamb * f

    print(f"lamb = {lamb} m\n"
          f"    f = {f} Hz\n"
          f"    v = {v} m / s")
```

```
lamb = 0.6 m
    f = 5.0 Hz
    v = 3.0 m / s
```

15. A wave on a string is described by the following function:

$$y(x, t) = (0.0050 \text{ m}) \cos \left[\left(4.0 \frac{\text{rad}}{\text{m}} \right) x - \left(12 \frac{\text{rad}}{\text{s}} \right) t \right]$$

What are the wave speed v and direction of travel?

The above equation describes a sinusoidal wave function moving in the $+x$ direction. The general form is as follows, with values annotated:

$$y(x, y) = A \cos(kx - \omega t) \quad (8)$$

$$\text{where:} \quad (9)$$

$$A = 0.0050 \text{ m} \quad (10)$$

$$k = 4.0 \frac{\text{rad}}{\text{m}} \quad (11)$$

$$x = ? \text{ m} \quad (12)$$

$$\omega = 12 \frac{\text{rad}}{\text{s}} \quad (13)$$

$$t = ? \text{ s} \quad (14)$$

$$(15)$$

```
[2]: A = 0.0050 ## m
    k = 4.0 ## rad / m
    omega = 12.0 ## rad / s
```

Given the angular frequency ω , we can solve for the frequency f .

$$\omega = 2\pi f \quad (16)$$

$$f = \frac{\omega}{2\pi} \quad (17)$$

$$= \frac{12 \frac{\text{rad}}{\text{s}}}{2\pi} \quad (18)$$

$$= 1.91 \frac{1}{\text{s}} \quad (19)$$

$$= 1.91 \text{ Hz} \quad (20)$$

We can also solve for the wavelength λ using the formula for the wave number k .

$$k = \frac{2\pi}{\lambda} \quad (21)$$

$$\lambda k = 2\pi \quad (22)$$

$$\lambda = \frac{2\pi}{k} \quad (23)$$

$$= \frac{2\pi}{4.0 \frac{\text{rad}}{\text{m}}} \quad (24)$$

$$= 1.57 \text{ m} \quad (25)$$

Finally, we can use the wavelength λ and frequency f to solve for the wave speed v .

$$v = \lambda f \quad (26)$$

$$= 1.57 \text{ m} \times 1.91 \frac{1}{\text{s}} \quad (27)$$

$$= 3.00 \frac{\text{m}}{\text{s}} \quad (28)$$

```
[7]: f = omega / math.tau
print(f"f = {f:.14g} Hz")

lamb = math.tau / k
print(f"lamb = {lamb:.14g} m")

v = lamb * f
print(f"v = {v} m / s")
```

```
f = 1.9098593171027 Hz
lamb = 1.5707963267949 m
v = 3.0 m / s
```

16. A sinusoidal wave on a string has a frequency $f = 50 \text{ Hz}$ and travels at $v = 20 \frac{\text{m}}{\text{s}}$. What is the wave number k ?

In order to determine the wave number k , we first need the wave length λ :

$$v = \lambda f \quad (29)$$

$$\lambda = \frac{v}{f} \quad (30)$$

$$\lambda = \frac{20 \frac{\text{m}}{\text{s}}}{50 \frac{1}{\text{s}}} \quad (31)$$

$$= 0.4 \text{ m} \quad (32)$$

Now we can determine the wave number.

$$k = \frac{2\pi}{\lambda} \quad (33)$$

$$= \frac{2\pi}{0.4 \text{ m}} \quad (34)$$

$$= 15.7 \frac{\text{rad}}{\text{m}} \quad (35)$$

```
[10]: f = 50.0 ## Hz
      v = 20.0 ## m / s
      lamb = v / f
      k = math.tau / lamb

      print(f"    f = {f} Hz\n"
            f"    v = {v} m / s\n"
            f"    lamb = {lamb} m\n"
            f"    k = {k:.14g} rad / m")
```

```
f = 50.0 Hz
v = 20.0 m / s
lamb = 0.4 m
k = 15.707963267949 rad / m
```

17. A string has a mass $m = 5.0 \text{ g}$ and a length $L = 2.0 \text{ m}$, and is under a tension $F = 80 \text{ N}$. What is the speed of transverse waves v on this string?

We need to calculate the linear mass density μ , which is the just the mass m , in kilograms, per unit length L , in meters.

$$\mu = \frac{m}{L} \quad (36)$$

$$= \frac{5.0 \text{ g} \times \frac{1 \text{ kg}}{1000 \text{ g}}}{2.0 \text{ m}} \quad (37)$$

$$= \frac{0.005 \text{ kg}}{2.0 \text{ m}} \quad (38)$$

$$= 0.0025 \frac{\text{kg}}{\text{m}} \quad (39)$$

Thus the speed of transverse waves v can be determined as follows:

$$v = \sqrt{\frac{F}{\mu}} \quad (40)$$

$$= \sqrt{\frac{80 \text{ N}}{0.0025 \frac{\text{kg}}{\text{m}}}} \quad (41)$$

$$= \sqrt{\frac{80 \text{ kg} \cdot \text{m} \cdot \text{s}^{-2}}{0.0025 \frac{\text{kg}}{\text{m}}}} \quad (42)$$

$$= \sqrt{\frac{80 \text{ kg} \cdot \text{m} \cdot \text{s}^{-2} \text{ m}}{0.0025 \text{ kg}}} \quad (43)$$

$$= \sqrt{\frac{80}{0.0025} \text{m}^2 \cdot \text{s}^{-2}} \quad (44)$$

$$= 179 \frac{\text{m}}{\text{s}} \quad (45)$$

18. The tension F in a string is increased by a factor of 9 and, at the same time, its linear mass density μ is increased by a factor of 4. BY what factor does the wave speed change?

$$v = \sqrt{\frac{9F}{4\mu}} \quad (46)$$

The factor in question is $\sqrt{\frac{9}{4}} = \frac{3}{2} = 1.5$.

19. On the same string with fixed tension, a wave's amplitude A is tripled and its frequency f is halved. By what factor does the average power P_{av} change?

$$P_{av} = \frac{1}{2} \sqrt{\mu F} \times \omega^2 \times 3A^2 \quad (47)$$

$$\text{since:} \quad (48)$$

$$\omega = 2\pi f \quad (49)$$

$$\text{thus:} \quad (50)$$

$$P_{av} = \frac{1}{2} \sqrt{\mu F} \times (2\pi f)^2 \times 3A^2 \quad (51)$$

$$P_{av} = \frac{1}{2} \sqrt{\mu F} \times (2\pi \frac{1}{2} f)^2 \times 9A \quad (52)$$

$$P_{av} = \frac{1}{2} \sqrt{\mu F} \times \frac{1}{2} (2\pi f)^2 \times 9A \quad (53)$$

$$P_{av} = \frac{1}{2} \sqrt{\mu F} \times \frac{1}{4} (2\pi f)^2 \times 9A \quad (54)$$

$$(55)$$

Thus the factor is $\frac{1}{4} \times 9 = 2.25$.

20. A small source radiates waves uniformly in all directions with no absorption. At radius $r = 2.0$ m from the source, the intensity $I = 8.0 \text{ W} \cdot \text{m}^{-2}$. What is the intensity at radius $x' = 6.0$ m?

This is an application of the *Inverse-square law*. The question is asking us how strong the intensity I is at a farther offset.

21. A string a length $L = 0.80$ m is fixed at both ends, and waves travel on it at $v = 160 \frac{\text{m}}{\text{s}}$. What is the fundamental frequency f_1 ?

$$\lambda_n = \frac{2L}{n} \quad (56)$$

$$\lambda_1 = \frac{20.80 \text{ m}}{1} \quad (57)$$

$$= 1.60 \text{ m} \quad (58)$$

$$(59)$$

$$v = \lambda f \quad (60)$$

$$f = \frac{v}{\lambda} \quad (61)$$

$$= \frac{160 \frac{\text{m}}{\text{s}}}{1.60 \text{ m}} \quad (62)$$

$$= 100 \frac{1}{\text{s}} \quad (63)$$

$$= 100 \text{ Hz} \quad (64)$$

22. For the same string, $L = 0.80$ m, $v = 160 \frac{\text{m}}{\text{s}}$, fixed at both ends, what are the frequency and wavelength of the third harmonic?

$$f_n = n \times f_1 \quad (65)$$

$$f_3 = 3 \times f_1 \quad (66)$$

$$= 3 \times 100 \text{ Hz} \quad (67)$$

$$= 300 \text{ Hz} \quad (68)$$

23. A guitar string has a vibrating length $L = 0.65$ m and a linear mass density $\mu = 4.0 \frac{\text{g}}{\text{m}}$. What tension F is required for a fundamental frequency $f_1 = 110$ Hz?

$$f_1 = \frac{1}{2L} \sqrt{\frac{F}{\mu}} \quad (69)$$

$$(f_1 = \frac{1}{2L} \frac{\sqrt{F}}{\sqrt{\mu}}) \frac{2L}{1} \quad (70)$$

$$(2Lf_1 = \frac{\sqrt{F}}{\sqrt{\mu}}) \frac{\sqrt{\mu}}{1} \quad (71)$$

$$(\sqrt{F} = 2Lf_1\sqrt{\mu})^2 \quad (72)$$

$$F = 4L^2 f_1^2 \mu \quad (73)$$

$$= 4 \times (0.65 \text{ m})^2 \times (110 \frac{1}{\text{s}})^2 \times 4.0 \frac{\text{g}}{\text{m}} (\frac{1 \text{ kg}}{1000 \text{ g}}) \quad (74)$$

$$= 82 \text{ kg} \cdot \text{m} \cdot \text{s}^{-2} \quad (75)$$

$$= 82 \text{ N} \quad (76)$$

24. The guitar string in the previous problem, with fundamental frequency $f_1 = 110$ Hz, vibrating length $L = 0.65$ m, is pressed against a fret that only $\lambda_n = 0.55$ m can vibrate. The tension and the string are unchanged. What is the new fundamental frequency?

Recall that we previously solved for the tension of the fundamental frequency $F_1 = 82$ N.

$$f_n = \frac{1}{2L} \sqrt{\frac{F}{\mu}} \quad (77)$$

$$= \frac{1}{2 \times 0.55 \text{ m}} \sqrt{\frac{82 \text{ kg} \cdot \text{m} \cdot \text{s}^{-2}}{0.004 \frac{\text{kg}}{\text{m}}}} \quad (78)$$

$$= 0.9091 \frac{1}{\text{m}} \times 143.18 \frac{\text{m}}{\text{s}} \quad (79)$$

$$= 130.16 \frac{1}{\text{s}} \quad (80)$$

$$= 130 \text{ Hz} \quad (81)$$

$$(82)$$

25. Two traveling waves, each of amplitude $A = 3.0$ mm and frequency $f = 120$ Hz, move in opposite directions along a string at speed $v = 48 \frac{\text{m}}{\text{s}}$, producing a standing wave. What is the amplitude at an antinode, and what is the distance between adjacent nodes?